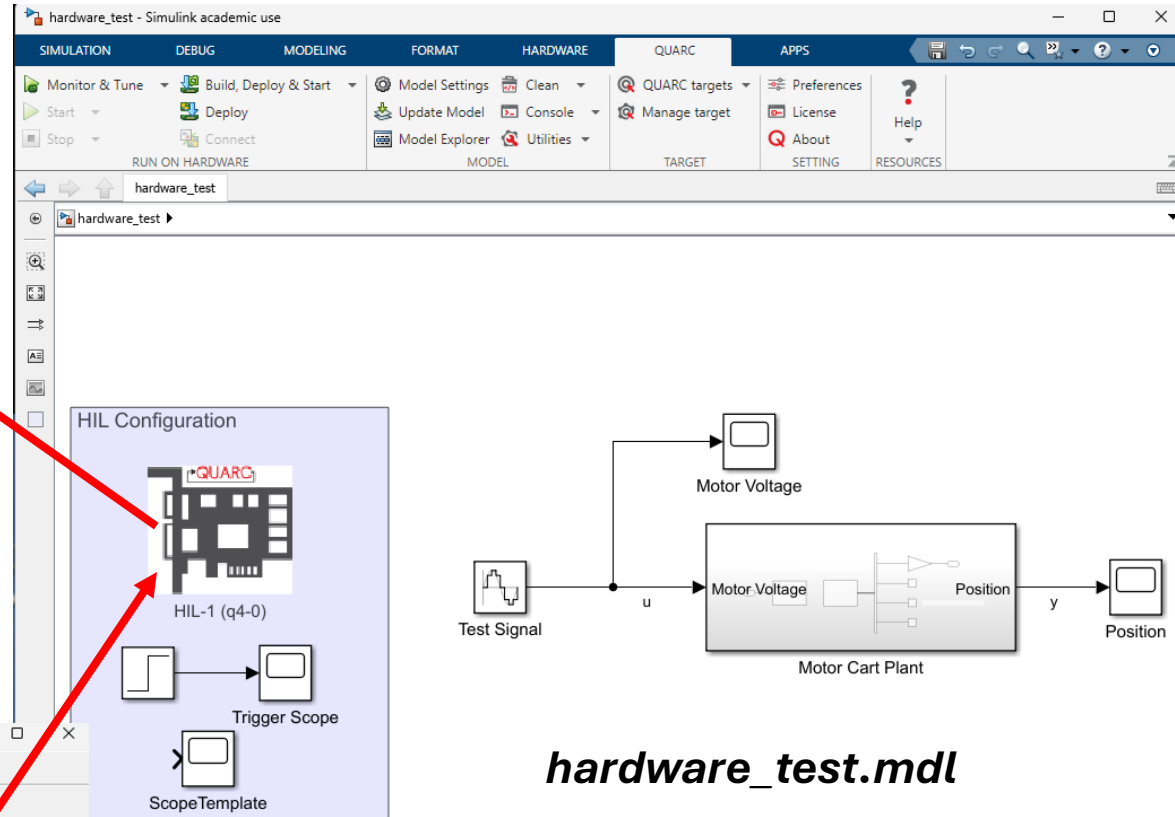
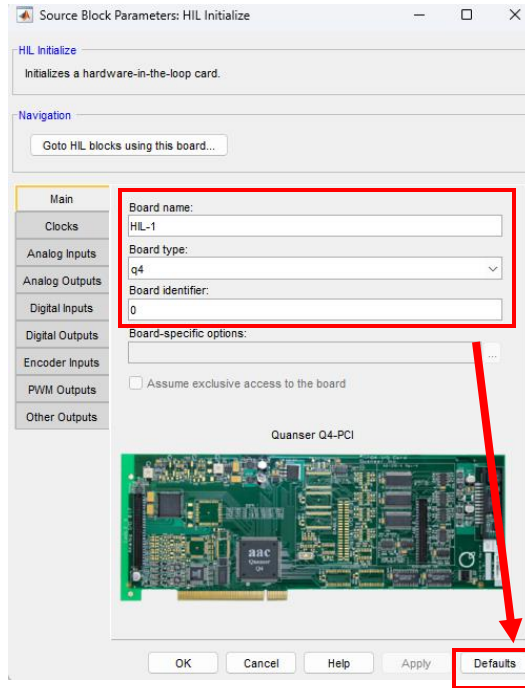
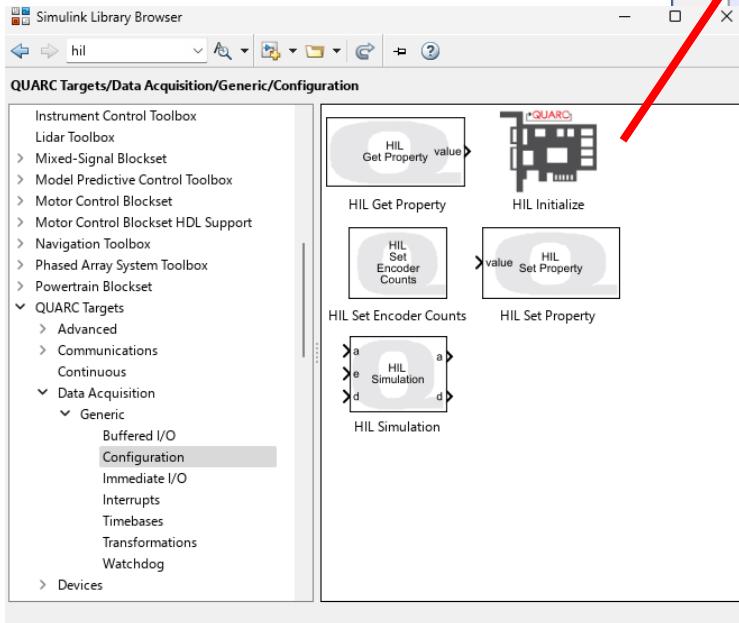


Configuring the model to work with DAQ

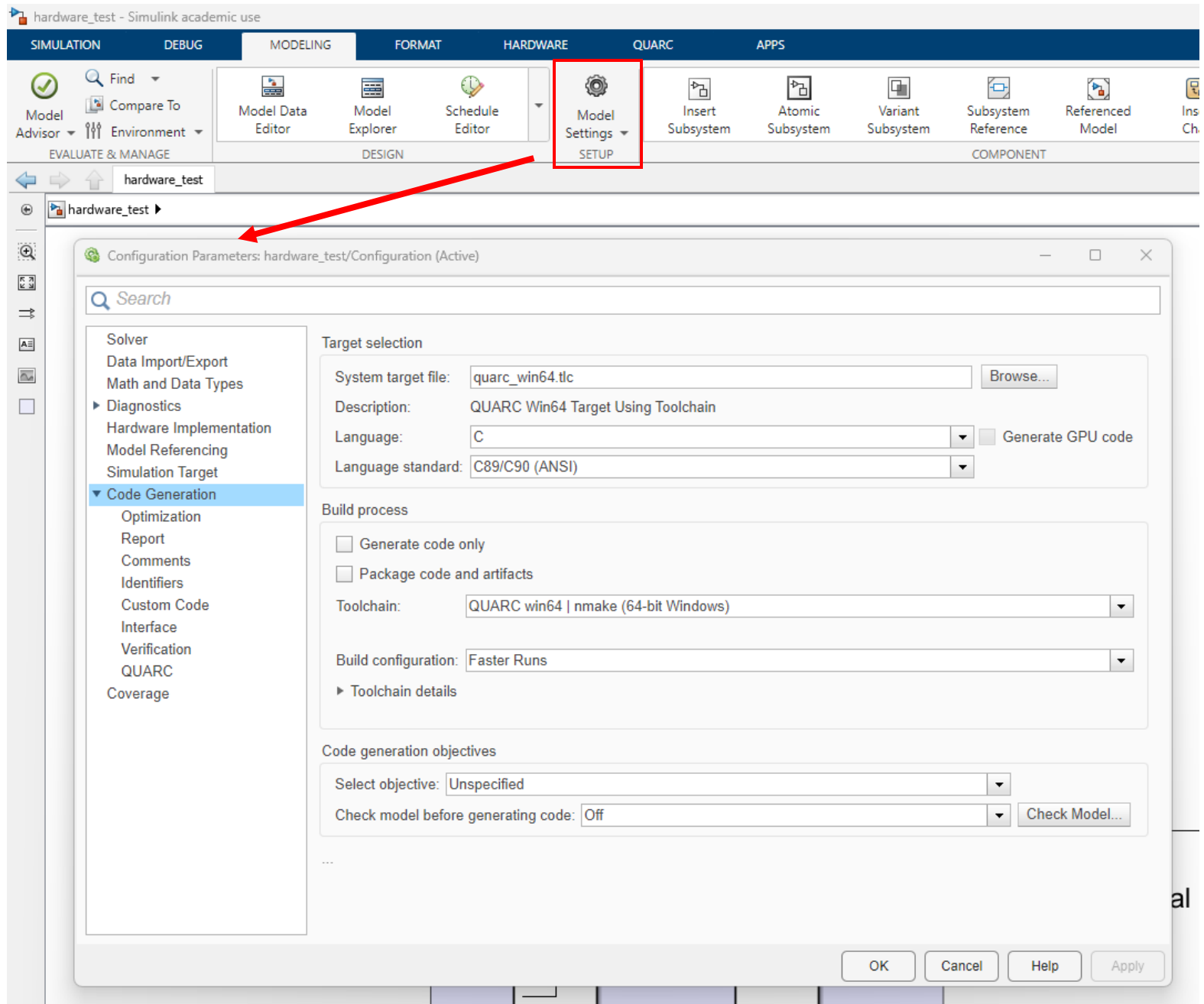


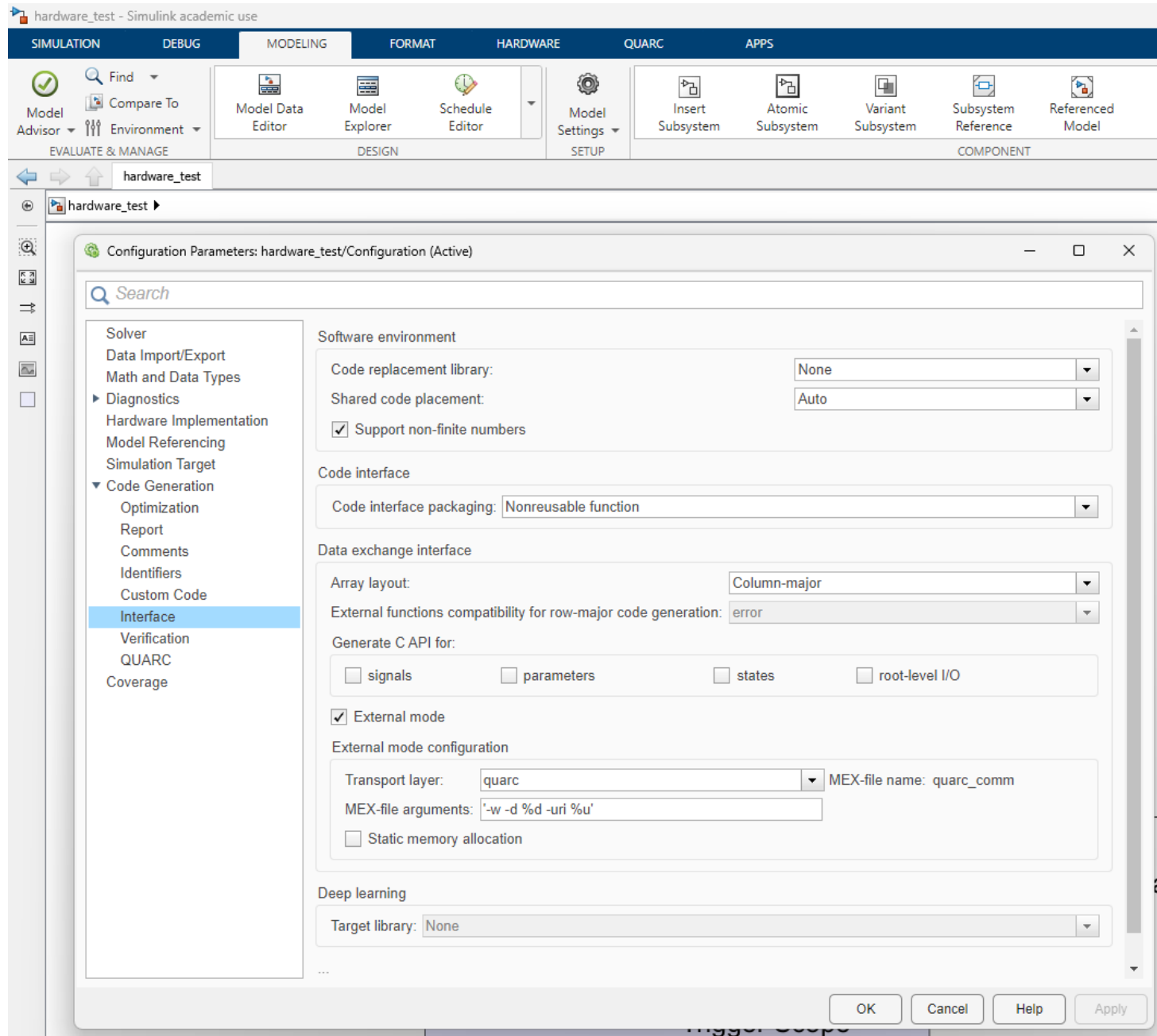
hardware_test.mdl

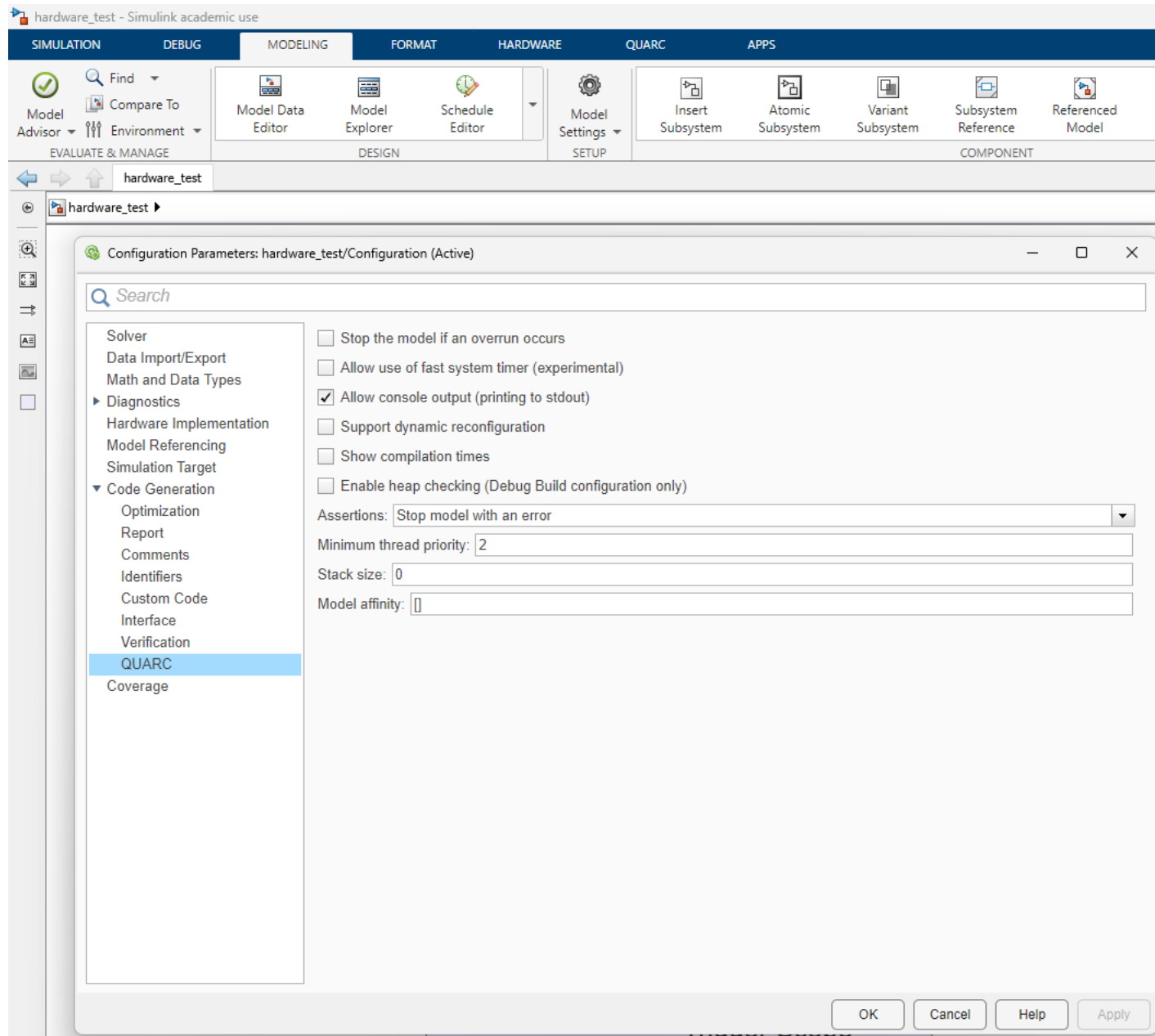


Set the HIL Initialize block to have board name HIL-1 and Board type q4. Click defaults to set up

Configuring model parameters for HIL & Quarc







Configuration Parameters: hardware_test/Configuration (Active)

Search

- Solver
- Data Import/Export
- Math and Data Types
- ▶ Diagnostics
- Hardware Implementation
- Model Referencing
- Simulation Target
- ▼ Code Generation
 - Optimization
 - Report
 - Comments
 - Identifiers
 - Custom Code
 - Interface
 - Verification
 - QUARC
 - Coverage

Simulation time

Start time: 0.0 Stop time: 10

Solver selection

Type: Fixed-step Solver: discrete (no continuous states)

▼ Solver details

Fixed-step size (fundamental sample time): 0.001

Tasking and sample time options

Periodic sample time constraint: Unconstrained

- ☒ Treat each discrete rate as a separate task
- ☐ Allow tasks to execute concurrently on target
- ☐ Automatically handle rate transition for data transfer
- ☐ Allow multiple tasks to access inputs and outputs
- ☒ Higher priority value indicates higher task priority

Important to have these set correctly
 $T_s = 0.001$, if you change it here change it everywhere...

OK Cancel Help Apply

Configuration Parameters: hardware_test/Configuration (Active)

Search

Solver

Data Import/Export

Math and Data Types

▶ Diagnostics

Hardware Implementation

Model Referencing

Simulation Target

▼ Code Generation

Optimization

Report

Comments

Identifiers

Custom Code

Interface

Verification

QUARC

Coverage

Hardware board: Determine by Code Generation system target file

Code Generation system target file: [quarc_win64.tlc](#)

Device vendor: Intel

Device type: x86-64 (Windows64)

▼ Device details

Number of bits

char: 8

short: 16

int: 32

long: 32

long long: 64

float: 32

double: 64

native: 64

pointer: 64

size_t: 64

ptrdiff_t: 64

Largest atomic size

integer: Char

floating-point: Float

Byte ordering: Little Endian

Signed integer division rounds to: Zero

☒ Shift right on a signed integer as arithmetic shift

☐ Support long long

OK

Cancel

Help

Apply

Configuring scopes for data logging:

The image shows a MATLAB/Simulink HIL Configuration window with a block diagram. A block labeled 'ScopeTemplate' is highlighted with a blue border. A red arrow points from this block to the 'Configuration Properties: ScopeTemplate' dialog box. Another red arrow points from the 'View' menu in the 'ScopeTemplate' window to the 'Configuration Properties ...' option. The dialog box has tabs for 'Main', 'Time', 'Display', and 'Logging'. The 'Logging' tab is selected, showing options for data logging. A third red arrow points from the text 'Saved as an array in the workspace' to the 'Save format' dropdown menu, which is set to 'Array'.

ScopeTemplate

Configuration Properties: ScopeTemplate

Logging

☐ Limit data points to last: 5000

☒ Decimation: 1

☒ Log data to workspace

Variable name: ScopeTemplate

Save format: Array

Variable name is what will appear in the workspace after the run is complete

OK Cancel Apply

ScopeTemplate(:,1) = 'time'

ScopeTemplate(:,2) = 'data'

Saved as an array in the workspace

Configuring triggering and model data logging:

The screenshot shows the Simulink hardware configuration interface. The 'HARDWARE' tab is selected in the top menu. The 'Control Panel' icon is highlighted with a red box. A red arrow points from this icon to the 'hardware_test: External Mode Control Panel' dialog. Another red arrow points from the 'Signal & Triggering ...' button in the dialog to the 'hardware_test: External Signal & Triggering' dialog.

hardware_test: External Mode Control Panel

Connection and triggering

Connect Start Real-Time Code Arm Trigger

Floating scope

☒ Enable data uploading

Duration: auto

Parameter tuning

☐ Batch download

Download

Configuration

Signal & Triggering ... Data Archiving ...

OK Help

hardware_test: External Signal & Triggering

Signal selection

Trigger	Selected	Block	Path
	X	Motor Voltage	hardware_test/Motor Voltage
	X	Position	hardware_test/Position
T	X	Trigger Scope	hardware_test/Trigger Scope

☒ Select all

Clear all

☐ on

☐ off

Trigger Signal

Go To Block

Trigger options

Source: signal Mode: normal Duration: 1e+06 Delay: 0

☒ Arm when connecting to target

Trigger signal

Path: hardware_test/Trigger Scope Port: 1 Element: any

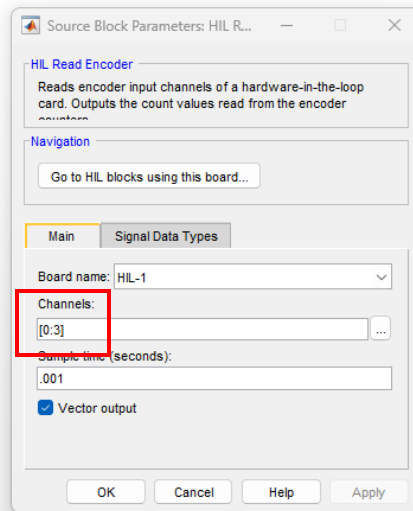
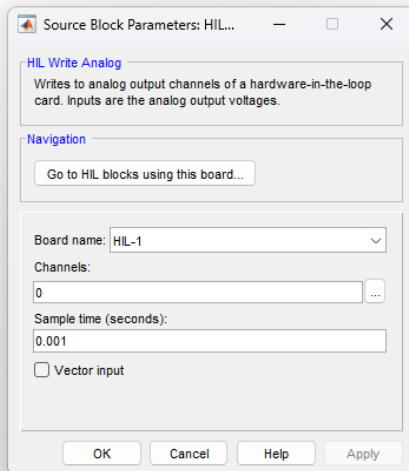
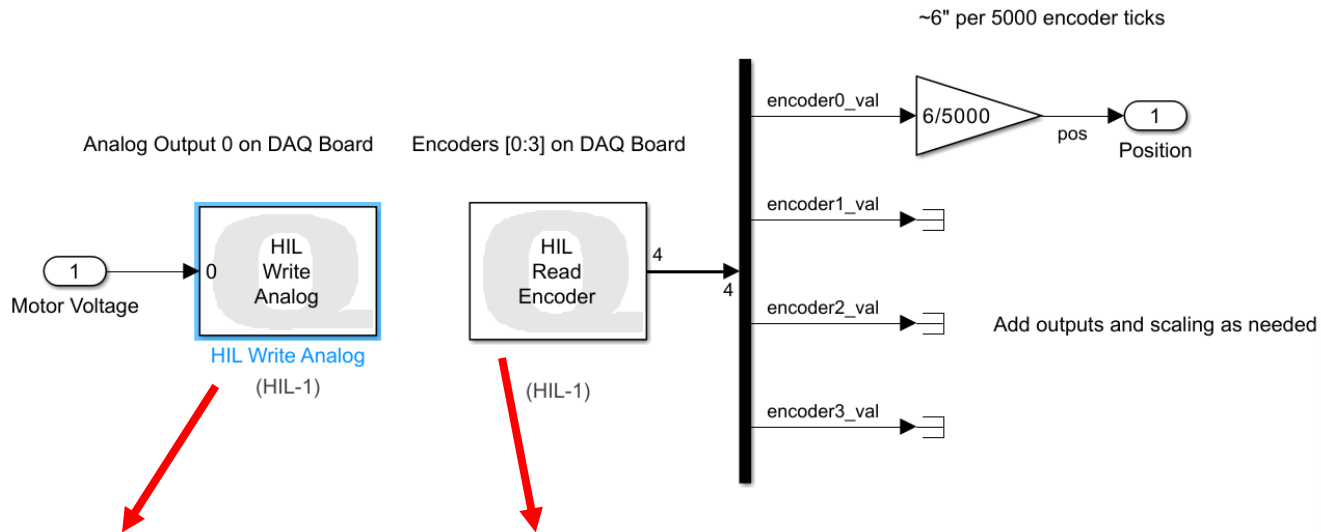
Direction: rising Level: 0.5 Hold-off: 0

OK Cancel Help Apply

X under Selected means the signal will be logged

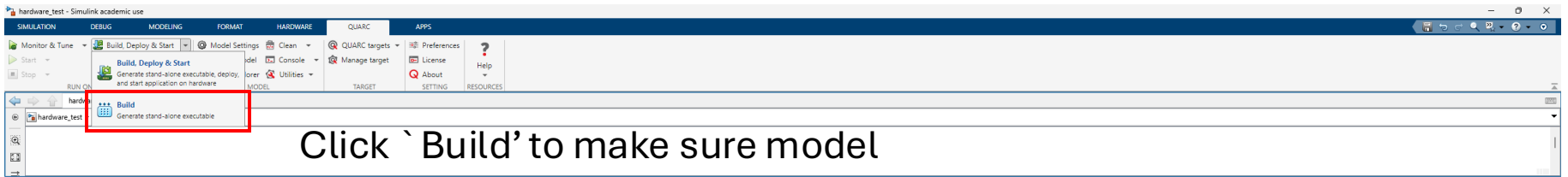
T under trigger means that will be used as trigger signal (don't change)

Duration is maximum # of samples logged, $Duration = t_{Stop}/T_s$



Demo model is reading all 4 encoders as vector and demuxxing (shown above)

Can set to just read a single encode



Click `Build` to make sure model
can be compiled

Diagnostic Viewer

2:26 PM: Build

1 of 1 models built (0 models already up to date)
Build duration: 0h 0m 9.4402s

Build Summary

Top model targets built:

Model	Action	Rebuild Reason
hardware_test	Code generated and compiled.	Generated code was out of date.

1 of 1 models built (0 models already up to date)
Build duration: 0h 0m 9.4402s

```
### Generating code and artifacts to "Model specific" folder structure
### Generating code into build folder: C:\Users\maxcrisfull\Downloads\hardware_test_quarc_wln64
### Invoking Target Language Compiler on hardware_test.rtw
### Using System Target File: C:\Program Files\Quanser\QUARC\quarc\targets\toolchain\wln64\quarc_wln64.tlc
### Loading TLC function libraries
.....
### Initial pass through model to cache user defined code
..
### Caching model source code
.....
### Writing header file hardware_test_types.h
### Writing header file hardware_test.h
### Writing header file rttypes.h
### Writing header file multiword_types.h
.
### Writing header file zero_crossing_types.h
### Writing source file hardware_test.c
### Writing header file hardware_test_private.h
### Writing header file rtmodel.h
### Writing source file hardware_test_data.c
### Writing source file rt_nonfinite.h
.
### Writing source file rt_nonfinite.c
### Writing header file rt_defines.h
### Writing header file rtsetInf.h
### Writing source file rtsetInf.c
### Writing header file rtsetNaN.h
.
### Writing source file rtsetNaN.c
### TLC code generation complete (took 1.723s).
### Saving binary information cache.
### Using toolchain: QUARC wln64 | make (64-bit Windows)
### Creating 'C:\Users\maxcrisfull\Downloads\hardware_test_quarc_wln64\hardware_test.mk' ...
### Building 'hardware_test': make -f hardware_test.mk all

C:\Users\maxcrisfull\Downloads\hardware_test_quarc_wln64\call "setup_msvc.bat"

C:\Users\maxcrisfull\Downloads\hardware_test_quarc_wln64\set "VSCMD_START_DIR=C:\Users\maxcrisfull\Downloads\hardware_test_quarc_wln64"

C:\Users\maxcrisfull\Downloads\hardware_test_quarc_wln64"C:\Program Files\Microsoft Visual Studio\2022\Community\VC\Auxiliary\Build\VCVARSALL.BAT" ^> AMD64
*****
** Visual Studio 2022 Developer Command Prompt v17.6.5
** Copyright (c) 2022 Microsoft Corporation
[vcvarsall.bat] Environment initialized for: 'x64'

Microsoft (R) Program Maintenance Utility Version 14.36.32537.0
Copyright (C) Microsoft Corporation. All rights reserved.

c1 -c -nologo -GS -W4 -DNIN32 -D_JIT -D_DLL -HD -DRTAPI1 -Dcdec1 -DRTAPI2 -Dcdec1 -D_JAV64_1 -DNIN64 -DNIN32 -D_JIN32 -D_JININT -D_JIN32_MINT -Dx502 -DNT001_VERSION -Dx5020000 -D_JIN32_IE -Dx0600 -DINVER -Dx502 -DVGWORKS -DQUARC -DTARGET_TYPE -wln64 /wd4100 /wd4996 -D_CRT_SECURE_NO_DEPRECATED -DRT_FILE -DONESTEPPCH -DTERPFCM1 -DMULTI_INSTANCE_CODE -DINTEGER_CODE -DINT -DITD01EQ -DHMODEL -Dhardware_test -DNHST1 -DNKSTATES -DHAVESTDIO -DRT -DUSE_RTHMODEL @hardware_test_comp.rsp -Fo"hardware_test.obj" "C:\Users\maxcrisfull\Downloads\hardware_test_quarc_wln64\hardware_test.c"
hardware_test.c
c1 -c -nologo -GS -W4 -DNIN32 -D_JIT -D_DLL -HD -DRTAPI1 -Dcdec1 -DRTAPI2 -Dcdec1 -D_JAV64_1 -DNIN64 -DNIN32 -D_JIN32 -D_JININT -D_JIN32_MINT -Dx502 -DNT001_VERSION -Dx5020000 -D_JIN32_IE -Dx0600 -DINVER -Dx502 -DVGWORKS -DQUARC -DTARGET_TYPE -wln64 /wd4100 /wd4996 -D_CRT_SECURE_NO_DEPRECATED -DRT_FILE -DONESTEPPCH -DTERPFCM1 -DMULTI_INSTANCE_CODE -DINTEGER_CODE -DINT -DITD01EQ -DHMODEL -Dhardware_test -DNHST1 -DNKSTATES -DHAVESTDIO -DRT -DUSE_RTHMODEL @hardware_test_comp.rsp -Fo"hardware_test_data.obj" "C:\Users\maxcrisfull\Downloads\hardware_test_quarc_wln64\hardware_test_data.c"
hardware_test_data.c
c1 -c -nologo -GS -W4 -DNIN32 -D_JIT -D_DLL -HD -DRTAPI1 -Dcdec1 -DRTAPI2 -Dcdec1 -D_JAV64_1 -DNIN64 -DNIN32 -D_JIN32 -D_JININT -D_JIN32_MINT -Dx502 -DNT001_VERSION -Dx5020000 -D_JIN32_IE -Dx0600 -DINVER -Dx502 -DVGWORKS -DQUARC -DTARGET_TYPE -wln64 /wd4100 /wd4996 -D_CRT_SECURE_NO_DEPRECATED -DRT_FILE -DONESTEPPCH -DTERPFCM1 -DMULTI_INSTANCE_CODE -DINTEGER_CODE -DINT -DITD01EQ -DHMODEL -Dhardware_test -DNHST1 -DNKSTATES -DHAVESTDIO -DRT -DUSE_RTHMODEL @hardware_test_comp.rsp -Fo"hardware_test_main.obj" "C:\Users\maxcrisfull\Downloads\hardware_test_quarc_wln64\hardware_test_main.c"
hardware_test_main.c
c1 -c -nologo -GS -W4 -DNIN32 -D_JIT -D_DLL -HD -DRTAPI1 -Dcdec1 -DRTAPI2 -Dcdec1 -D_JAV64_1 -DNIN64 -DNIN32 -D_JIN32 -D_JININT -D_JIN32_MINT -Dx502 -DNT001_VERSION -Dx5020000 -D_JIN32_IE -Dx0600 -DINVER -Dx502 -DVGWORKS -DQUARC -DTARGET_TYPE -wln64 /wd4100 /wd4996 -D_CRT_SECURE_NO_DEPRECATED -DRT_FILE -DONESTEPPCH -DTERPFCM1 -DMULTI_INSTANCE_CODE -DINTEGER_CODE -DINT -DITD01EQ -DHMODEL -Dhardware_test -DNHST1 -DNKSTATES -DHAVESTDIO -DRT -DUSE_RTHMODEL @hardware_test_comp.rsp -Fo"rtsetInf.obj" "C:\Users\maxcrisfull\Downloads\hardware_test_quarc_wln64\rtsetInf.c"
rtsetInf.c
c1 -c -nologo -GS -W4 -DNIN32 -D_JIT -D_DLL -HD -DRTAPI1 -Dcdec1 -DRTAPI2 -Dcdec1 -D_JAV64_1 -DNIN64 -DNIN32 -D_JIN32 -D_JININT -D_JIN32_MINT -Dx502 -DNT001_VERSION -Dx5020000 -D_JIN32_IE -Dx0600 -DINVER -Dx502 -DVGWORKS -DQUARC -DTARGET_TYPE -wln64 /wd4100 /wd4996 -D_CRT_SECURE_NO_DEPRECATED -DRT_FILE -DONESTEPPCH -DTERPFCM1 -DMULTI_INSTANCE_CODE -DINTEGER_CODE -DINT -DITD01EQ -DHMODEL -Dhardware_test -DNHST1 -DNKSTATES -DHAVESTDIO -DRT -DUSE_RTHMODEL @hardware_test_comp.rsp -Fo"rtsetNaN.obj" "C:\Users\maxcrisfull\Downloads\hardware_test_quarc_wln64\rtsetNaN.c"
rtsetNaN.c
```

View diagnostics

Monitor & Tune builds & runs the code.

Expected outputs shown

